ModelIt!

Student Activity Pack

Plate Tectonics: Predicting Where Earth Will Crack Next

Using Models to Forecast Earthquakes and Volcanic Activity

Grade 06L2 | NGSS MS-ESS2-2, MS-ESS2-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think earthquakes happen more in California than in Florida?

2 Do earthquakes and volcanoes always happen together? Why or why not?

3 If you could measure the stress building in the ground, could you predict an earthquake? Why might it be hard?

4 Draw what you think happens underground when two tectonic plates push into each other.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Mantle Convection Speed | Plate Boundary Type | Tectonic Stress | Earthquake Frequency | Volcanic Activity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fast Convergent

High convection, convergent boundary

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Slow Transform

Low convection, transform boundary

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Stress Cycle

Moderate convection, observe over time

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Plate Thickness — Thicker continental plates resist subduction while thinner oceanic plates dive underneath, affecting boundary behavior

Magma Viscosity — Thicker magma creates explosive volcanoes while thinner magma creates gentle flows, depending on silica content

Fault Depth — Shallow earthquakes are more destructive at the surface than deep earthquakes of the same magnitude

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If we know where tectonic plates meet, why can't we predict exactly when the next big earthquake will hit? Use evidence from your simulation to support your answers.

1. **Why do convergent boundaries have volcanoes but transform boundaries do not?**

2. **If earthquakes release stress, why do more earthquakes keep happening at the same location?**

3. **How is the stress-release cycle at plate boundaries similar to stretching a rubber band?**

4. **Why is predicting WHEN an earthquake will happen so much harder than predicting WHERE?**

5. **How does your model help explain why the Ring of Fire has 75% of the world's volcanoes?**

8 STEM Challenge: Design an Earthquake Early Warning System

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YOUR ENGINEERING MISSION

Design a community early warning and response system for a city located near two different types of plate boundaries.

THE SCENARIO

A coastal city sits near both a convergent boundary (offshore subduction zone) and a transform fault. The city council needs your team to design warning systems, building codes, and emergency procedures based on your model predictions of earthquake and volcanic risks.

Engineering Design Thinking

1. WHAT SENSOR SYSTEMS WILL DETECT EARLY SEISMIC ACTIVITY AND HOW MUCH WARNING TIME CAN THEY PROVIDE?
2. HOW WILL YOUR BUILDING CODES DIFFER FOR AREAS NEAR THE TRANSFORM FAULT VS. NEAR THE SUBDUCTION ZONE?
3. WHAT EMERGENCY COMMUNICATION SYSTEMS WILL REACH ALL RESIDENTS QUICKLY?
4. HOW WILL YOU DESIGN INFRASTRUCTURE (BRIDGES, HOSPITALS, SCHOOLS) TO WITHSTAND THE HAZARDS YOUR MODEL PREDICTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Seismologists use network sensor data and computational models to study earthquake patterns and improve prediction methods. Volcanologists monitor active volcanoes to forecast eruptions and protect communities. They earn $65,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Why do convergent plate boundaries produce volcanoes but transform boundaries do not?

- **A.** Transform boundaries are not hot enough to melt rock
- **B.** At convergent boundaries, one plate subducts into the mantle and melts, creating magma that rises as volcanoes. Transform boundaries have no subduction.
- **C.** Volcanoes only form near oceans, and transform boundaries are always inland
- **D.** Transform boundaries move too slowly to generate enough heat for volcanoes

2 In the plate tectonics model, tectonic stress builds and then drops repeatedly. What creates this cyclical pattern?

- **A.** The Sun's gravity periodically pulls on the plates, releasing stress
- **B.** Mantle convection speeds up and slows down on a regular schedule
- **C.** Earthquakes release accumulated stress (negative feedback), then convection rebuilds the stress, creating a repeating cycle
- **D.** Volcanic eruptions reset the stress to zero every few years

3 A region on a plate boundary has not experienced an earthquake in 200 years, while a nearby boundary has frequent small earthquakes. Which region poses a greater risk of a large earthquake?

- **A.** The region with frequent earthquakes, because it is clearly more active
- **B.** Neither region, because earthquakes are completely random
- **C.** The quiet region, because 200 years of stress accumulation without release means enormous energy is stored
- **D.** Both regions have exactly equal risk at all times

4 How does increasing mantle convection speed affect geological activity at a plate boundary?

- **A.** Faster convection has no effect on surface activity because the mantle is too deep
- **B.** Faster convection cools the plates, reducing both earthquakes and volcanoes
- **C.** Faster convection pushes plates harder, building stress faster and increasing earthquake frequency and volcanic activity
- **D.** Faster convection only affects divergent boundaries, not convergent or transform ones

5 **Scientists know California will have a major earthquake, but they cannot predict the exact date. What does the plate tectonics model reveal about why precise timing is impossible?**

A. Scientists do not have enough instruments to measure plate movement

B. The exact moment when accumulated stress overcomes rock friction depends on countless microscopic variables that are impossible to measure

C. Earthquakes happen randomly and follow no patterns at all

D. Plate tectonics theory is still unproven and may be wrong about California